

◆ Fashion-MNIST is a controlled test of spatial structure

Question: how much of a CNN's advantage comes from the spatial structure of the images?

The data

70,000 images, 28×28 , single-channel 8-bit grayscale

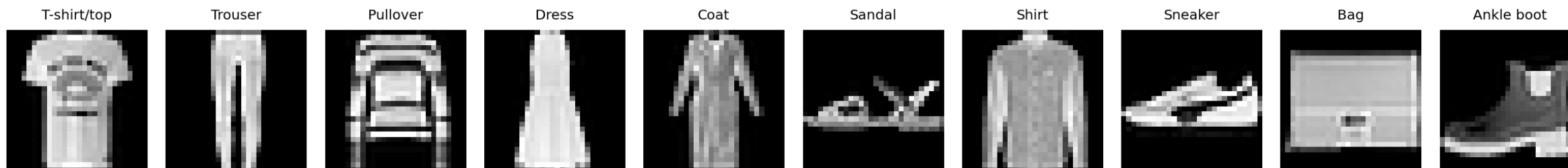
10 balanced clothing classes

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Official files, downloaded 22 Sep 2026

Why we chose it

- **Shared modelling question:** digital-human generation and deepfake-video detection both depend on spatial representations in vision models.
- **Suitable data property:** classes differ by silhouettes and local parts, so spatial arrangement should matter.
- **Clear but non-trivial:** accuracy spans 67.7–91.3 %, enough to separate methods.
- **Controlled and feasible:** one channel, 28×28 : little colour or background confounding; 30 fits are practical.



Team: Yingfeng Wang 700049354 · Chenye Yang 700052938 · Tianyi Wang 700053286

GitHub: https://github.com/cvlab-ycy/CSBP711_Assignment1

❖ We removed 23 development images to keep detected duplicate groups disjoint

Splits

Split	Images	Per class
Training	53,979	~5,400
Validation	5,998	~600
Test (official)	10,000	1,000

Same preprocessing for every model:

pixels ÷ 255; stratified 90/10 split saved with seed 711

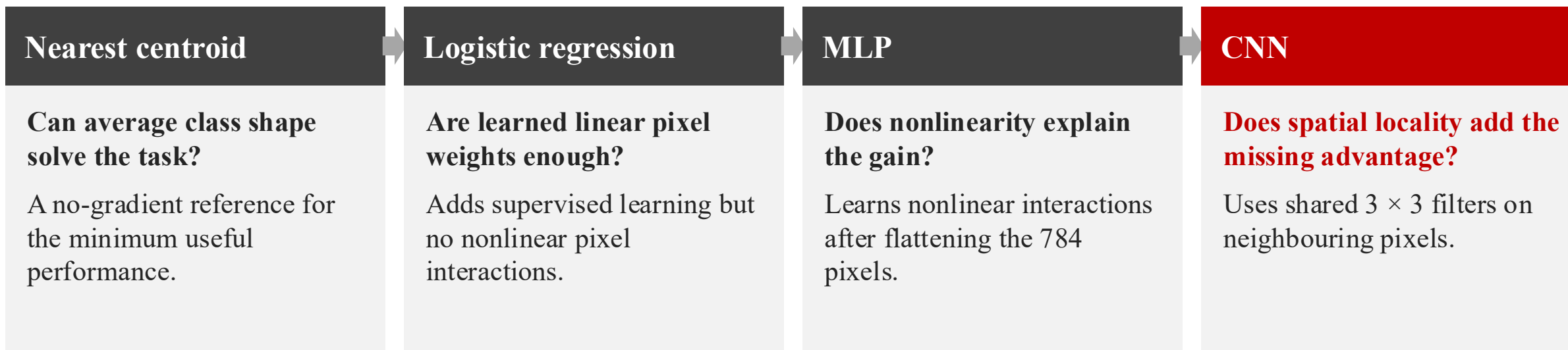
Problems found

0	missing values or exact repeats
31	detected near-duplicate pairs
23	development images removed
6 + 17	6 in groups touching test; 17 repeated development members

Why: near-copies across splits could inflate test accuracy. We removed development images from overlapping groups while keeping the official test set unchanged.

Limitation: the corrected audit followed inspection of the first results, so the final test is not a newly blind holdout.

Four models test increasingly structured assumptions



Why these four: centroid tests class averages; logistic adds learned linear weights; MLP adds nonlinearity; CNN adds local shared filters. The MLP–CNN comparison therefore tests whether locality explains the remaining accuracy gain.

Fair comparison: same split, preprocessing and seeds; validation-only learning-rate selection; no augmentation or pretrained weights.

◆ The CNN gains 3.0 points over the MLP with fewer parameters

Model	Test accuracy %	Training time (s)	Parameters
Nearest centroid	67.7 ± 0.0	0.02	0 (7,840 stored)
Logistic regression	84.4 ± 0.1	3.2	7,850
MLP	88.3 ± 0.2	6.8	118,282
CNN	91.3 ± 0.3	19.1	105,866

Trade-off: the CNN improves accuracy by 3.0 pp over the MLP with about 10 % fewer parameters, but takes 2.8× longer to train, so the gain has a compute cost. Parameter count alone therefore does not explain the CNN's advantage.

Mean $\pm$ sample SD across 3 seeds · same NVIDIA A100 GPU · training time includes early-stopping validation

Pixel shuffling reverses the CNN–MLP ranking

Data property

Clothing classes differ through local contours and parts: sleeves, soles, handles.

Why the CNN benefits

Shared 3×3 filters detect the same local pattern anywhere. The MLP has no shared neighbourhood detector.

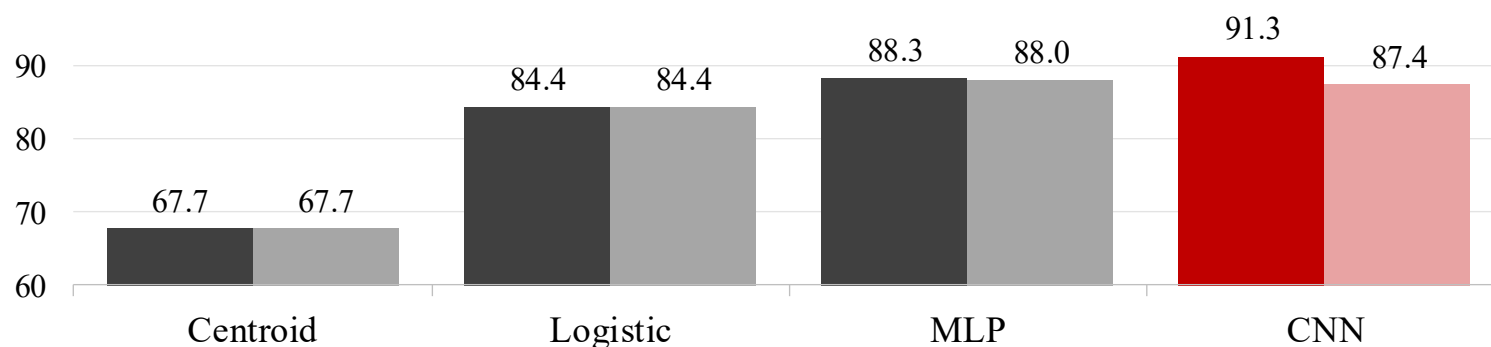
Controlled ablation

One fixed pixel shuffle for train, validation and test; every model retrained. Values and labels kept; true neighbours gone.

Prediction

The CNN should lose far more than the dense models, shrinking or reversing its lead over the MLP.

Test accuracy (%): dark = original, light = pixels shuffled



+3.03 → -0.63

CNN – MLP gap (pp), original → shuffled

Selective effect (accuracy change)

Centroid 0.00 · Logistic 0.00

MLP -0.23 · **CNN -3.88 pp**

Conclusion: the fixed shuffle preserves pixel values and labels but destroys neighbourhood structure. The dense controls stay stable while the CNN loses 3.88 points, reversing the CNN–MLP ranking. This selective drop supports spatial locality as the explanation under this protocol.

Scope: one dataset, one permutation and three training seeds.